

SGALL: SinGAN and YOLOv8 generated Acute Lymphoblastic Leukemia dataset for enhanced Detection and Classification

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Abstract—Early detection of Acute Lymphoblastic Leukaemia (ALL) using deep learning-enabled CAD models (DL-CAD) outperforms the conventional models (slow and prone to inter-personal observability). Development of a generalizable clinical utility DL-CAD model firmly depends on the size and variability of the dataset used for the training and validation process. GANs solve the data scarcity problem by synthetically generating diversified realistic samples. However, affirming the realism of generated samples is essential. Their use without appropriate validation introduces noise in the training data and potentially misguides the entire learning process. Moreover, training of GANs is computationally intensive and requires significant processing power and time. To address these bottlenecks, we propose a DL-CAD model that enables (i) generation of a diversified ALL dataset, namely SGALL, by leveraging the power of the lightweight generative model SinGAN and the YOLOv8 algorithm and (ii) classification of ALL samples using different pre-trained models. SinGAN, is trained on a single blood smear image through multi-scale pyramidal structure and has the capability to add variability in the generated dataset. YOLOv8 algorithm annotates the generated dataset to validate authentic samples and discards the ambiguous samples. Through extensive experimentation, performance analysis of different pre-trained classification models is verified on ALL samples generated through conventional data augmentation techniques, only SinGAN technique, and Annotated-SinGAN techniques (SGALL dataset). The DenseNet169 model trained on the SGALL dataset achieved 99.98% accuracy and 100% precision in diagnosing ALL samples.

Index Terms—ALL, DL-CAD, SinGAN, YOLOv8, SGALL

I. INTRODUCTION

Acute Lymphoblastic leukaemia (ALL) is a critical haematological malignancy that requires accurate and timely diagnosis for effective patient treatment [1]. The detection process involves identifying and classifying abnormal lymphoblast cells in peripheral blood smear images. Conventionally, leukaemia detection relied on the manual examination of blood smear slides by experienced haematologists. This process involved the visual inspection and identification of abnormal leukocytes under a microscope, using domain expertise to assess morphological features such as cell size, shape, nucleus-to-cytoplasm ratio, and staining

characteristics. This approach is time-consuming, subjective, and prone to inter-person errors [1]. CNN-based deep learning (DL) techniques have revolutionised leukaemia detection by automating the feature extraction and classification process [1]–[4]. These models require large amounts of data to learn deep features that enable them to distinguish between normal and abnormal cells with high precision. An efficient lymphoblastic leukaemia detection model must be trained on a large, balanced and well-labelled dataset of peripheral blood smear images to facilitate unbiased learning [3]. Most state-of-the-art datasets are private and require permission to use. Moreover, publicly available datasets lack a variety of blast cell morphologies and are therefore less diverse. Consequently, the model trained on these datasets shows limited generalization capability when validated on real-time unseen test samples [3]. To address the challenge of limited datasets, techniques such as data augmentation, transfer learning [3], and GANs [5]–[8] are explored. Traditional data augmentation techniques utilise predefined transformations such as translation, rotation, flipping, and brightness adjustment on existing data samples to boost the dataset volume, but cannot introduce newness to the data samples. In contrast, transfer learning based models utilise pre-trained models (trained on large datasets like ImageNet) and final classification layers are fine-tuned for classifications. However, fine-tuning pre-trained models on small ALL datasets can lead to overfitting. Among all, GANs are the most prevailing alternatives in enhancing the generalisation power of DL models. GANs generate different possible realistic output samples from the same latent input and thus boost the volume of existing datasets. GANs able to add diversity in the dataset by injecting noise vectors at a cost of heavy computational overload [7]. Moreover, the generated data samples from GANs need to be carefully validated before their further use in any downstream ALL classification process [9]. This paper is suggested to address these issues of GAN.

The contributions of the paper are:

- 1) Utilisation of SinGAN [10], a light-weight GAN, to generate synthetic images. The SinGAN network leverages GAN synthesis. It is trained on a single blood smear image using a multi-scale pyramidal structure to introduce new structures and configurations into synthetic images.
- 2) Annotation/labelling of SinGAN-generated data samples using YOLOv8 algorithm. The annotation process labels the domain-specific consistent samples and discards the ambiguous or non-realistic generated samples. The annotated and validated SinGAN-generated ALL data samples constitute the SGALL dataset. This process helps to overcome the limited availability of annotated data and validate the realism of SinGAN-generated samples.
- 3) Training and validation of several pre-trained models, such as VGGNets, ResNets, InceptionNets, DenseNets, EfficientNets and NasNet for the classification of ALL samples.

The entire paper is represented as follows: Section 2 outlines the existing state-of-the-art GAN-based DL methods for ALL diagnosis. Our proposed work is discussed in section 3. Result and analysis of the proposed work is discussed in section 4. Section 5 concludes the paper with some future scope of the research.

II. BACKGROUND STUDY

This section briefly illustrates the existing state-of-the-art GAN-based DL methodologies used for ALL detection and classification. Data scarcity and class imbalance are two major challenges for achieving an accurate deep learning-based ALL diagnosis model. GANs synthetically produce realistic images of ALL cells, increase dataset volume, and correct dataset imbalance. This improvement adds a significant improvement in developing accurate AI models for ALL diagnosis. Some existing GAN-integrated AI models for improved detection and classification of ALL samples are discussed below.

Authors in [5] used GAN with transfer learning models like DarkNet-53, ShuffleNet and achieved 99.70 % accuracy for the LISC dataset. Authors in [6] compared the performance of Deep Convolutional GAN (DCGAN) with three data augmentation techniques and extensively proved that DCGAN outperforms the classification accuracy. Authors in [7] utilised SimpleGAN, Wasserstein GAN and DCGAN architecture to enhance the volume of the dataset to deal with the dataset imbalance problem. DCGAN outperform other models. In [8], authors used GAN to enhance the volume of the dataset to enhance the classification accuracy. However, a limited set of marker combinations necessitates investigation of additional combinations to enhance classification performance. Authors in [11] used CNN for feature extraction and classification of blood smear images, whereas GANs are used as an augmentation module to deal with the class imbalance problem. Authors in [12] utilised a pipeline through the combination of Mayfly Optimisation (MO) algorithm (for efficient feature extraction), GANs (for data augmentation), followed by PCA (for dimensionality reduction) for leukaemia classification. The method achieved 99.8% classification accuracy. However,

integration of the MO algorithm with GAN and PCA makes the system computationally expensive, leading to its limited use in clinical practices. Authors in [13] used WGAN for the transformation of random noise vectors into low-resolution images, followed by the utilisation of image-to-image translation to convert the low-resolution images into high-resolution final images. Afterwards, CNN classifiers trained with these synthetic images provided accuracy from 87.7% to 100%. Along with these remarkable advantages, GAN model suffers from some major limitations such as:

- Checking the the realism of synthetic images generated through GAN is crucial. Utilising synthetic images without appropriate validation introduces noise in the training data. It potentially misguides the classifier and affects the generalisation capability of the model.
- Training of GANs is computationally intensive, requires significant processing power and time.
- Model performance is evaluated on a specific dataset, and its effectiveness needs to be validated with other datasets.

III. PROPOSED METHODOLOGY:

This section describes the proposed design methodology. Steps of the proposed work are summarised as follows:

- 1) Data collection, pre-processing and data augmentation.
- 2) Utilisation of SinGAN to generate synthetic ALL images.
- 3) Integration of YOLOv8 algorithm to annotate the SinGAN-generated data samples by measuring the percentage probabilities of blast and normal cells. The annotation process annotates the domain-specific consistent samples and discards the ambiguous or non-realistic samples generated to generate the SGALL dataset.
- 4) Training, validation and performance evaluation of pre-trained models when tested on the original dataset, the dataset with traditional augmentation techniques, the dataset generated from SinGAN and the SGALL dataset.

Figure 1 depicts the block diagram of the proposed work.

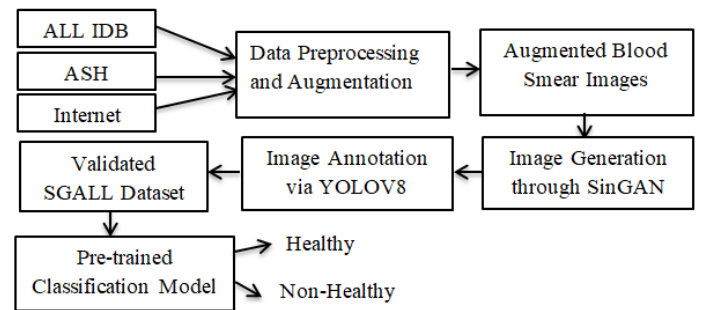


Fig. 1: Block diagram of the proposed work

A. Data Collection, Pre-processing and Data Augmentation

Initially, ALL blood smear images are collected from publicly available ALL IDB dataset [14], ASH (American Society of Hematology) database and some images are collected from

internet to generate new synthesized images using the SinGAN architecture. After resizing and image normalization, the conventional data augmentation techniques such as horizontal and vertical flips, rotation by various degrees, brightness manipulation, zooming, random translation and adding noise are introduced to expand the size of training samples.

B. SinGAN for generation of synthetic ALL samples

Conventional generative models rely on large training datasets to generate synthetic samples [6], whereas SinGAN generates new samples from a single training sample yet preserves variability in the generated dataset. SinGAN is an unconditional generative model that captures patch distribution at multiple scales (from coarse to fine) through a pyramid of multi-scale lightweight fully-convolutional GAN to generate synthetic images sequentially [10]. SinGAN injects noise at each level. The image generation process at level $s-1$ incorporates all generators $Gen_1...Gen_N$ and all random noise maps $z_1, ..., z_N$ up to this level. At each scale s , the model learns inherent patterns present in the previous coarse-scale image $\hat{I}_{s-1}$; up-samples, combines the injected noise map z_s with it and is fed to the Generator Gen_s to yield output synthetic image $\hat{I}_s$ through adversarial training. The entire image generation process of a generator Gen_s is described in equation (1).

$$\hat{I}_s = (\hat{I}_{s-1})^{\uparrow x} + \psi_s(z_s + ((\hat{I}_{s-1})^{\uparrow x})) \quad (1)$$

where, x is the up-sampling factor and ψ_s is the 5 layered convolutional blocks of the form Conv(3×3)-BatchNorm-LeakyReLU. Despite the limited receptive field, SinGAN can add much variability and global structure to the generated samples. Figure 2 depicts the generated samples using SinGAN by using raw samples from ASH, ALL IDB datasets.

C. Integration of YOLOv8 algorithm to prepare Labelled SGALL Dataset

This step uses the YOLOv8 algorithm to annotate the generated samples with their associated class labels. For this purpose, the YOLOv8 algorithm is trained on annotated ALL IDB and ASH datasets. The state-of-the-art benchmark ALL IDB and ASH datasets are annotated by expert haematologists, and the centroid coordinates of potential ALL lymphoblasts are also provided. These centroids are plotted on the respective images, and the blast cells are identified by visual inspection. Furthermore, Makesense.ai tool [15] is utilized to annotate the label of blast cells. Once the annotations are complete, YOLOv8 model [16] is trained on these annotated datasets to detect and localize blast and other normal cells in the generated dataset. Each image in the generated dataset undergoes validation using the YOLOv8 algorithm. The YOLOv8 model outputs the bounding box coordinates, class labels, and confidence scores for each detected cell in the image. During the validation process, if YOLOv8 model detects blast cells in an image, it is categorized as an unhealthy lymphoblastic sample. Conversely, if no blast cells are detected, the image is classified as a healthy cell. However, in certain cases,

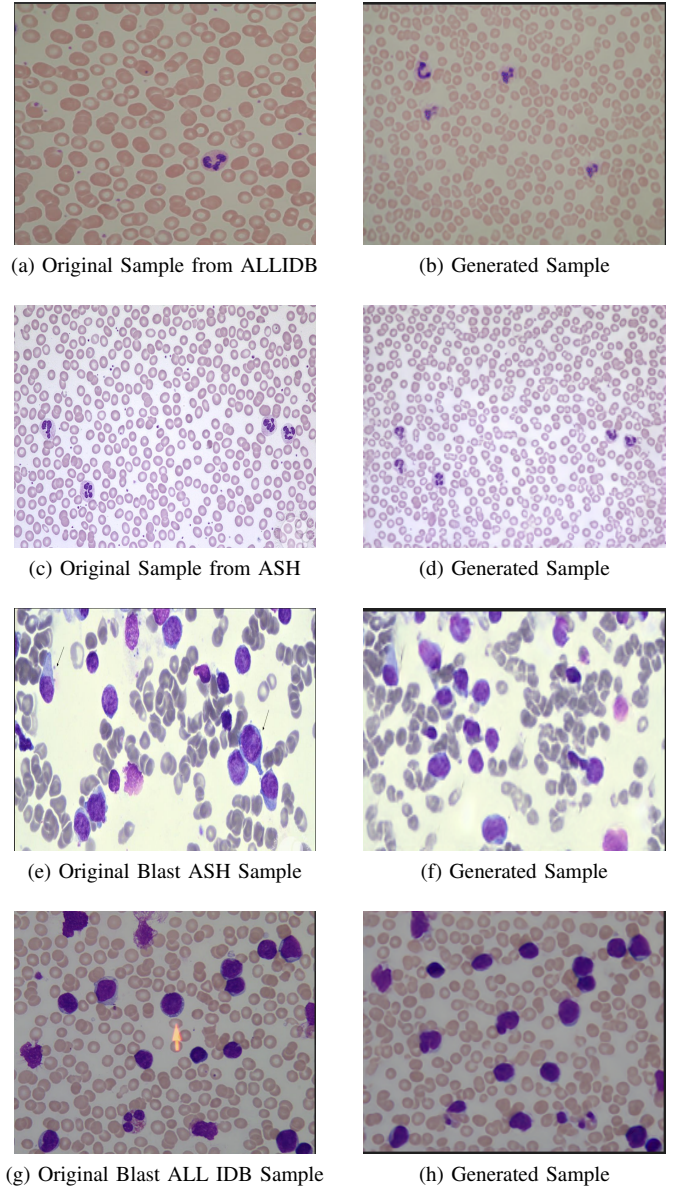


Fig. 2: Generated Samples using SinGAN

some images that are generated from healthy images are marked as unhealthy by YOLOv8 model and vice versa. Such ambiguous images are discarded from the dataset. Rigorous experiments have been performed and are cross-validated by Haematologists to enhance the quality and integrity of the dataset. Figure 3 depicts the entire annotation process using YOLOv8 algorithm. Figure 4 depicts the validation of SinGAN generated samples using YOLOv8 algorithm. Table I represents the details of the labelled and validated dataset (SGALL).

D. Transfer learning-based ALL classification

This section briefly describes the integration of six state-of-the-art transfer learning models i.e., VGGNets, ResNets, InceptionNets, DenseNets, EfficientNets and NasNet for the

TABLE III: Comparison of accuracy and loss (Training and Validation) values of pretrained models on All IDB Dataset + Augmentation

Model	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
VGG16	0.2198	0.9067	0.2633	0.9032
VGG19	0.2198	0.9333	0.2263	0.9032
InceptionV3	0.3195	0.9067	0.3658	0.8710
Resnet50	0.1442	0.9600	0.1882	0.9677
DenseNet121	0.2740	0.9470	0.2580	0.9680
DenseNet169	0.1671	0.9889	0.2492	0.9682
DenseNet201	0.2040	0.9735	0.2580	0.9679
EfficientNetB1	0.4944	0.8800	0.4617	0.8065
NasNet	0.2761	0.9200	0.2847	0.9032

TABLE IV: Comparison of accuracy and loss (Training and Validation) values of pretrained models on generated SGALL dataset

Model	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
VGG16	0.0222	0.9961	0.0199	0.9972
VGG19	0.0183	0.9971	0.0171	0.9977
InceptionV3	0.0323	0.9922	0.0250	0.9932
Resnet50	0.0209	0.9932	0.0216	0.9937
DenseNet121	0.0131	0.9976	0.0129	0.9977
DenseNet169	0.0038	0.9995	0.0031	0.9994
DenseNet201	0.0144	0.9978	0.0107	0.9990
EfficientNetB1	0.0530	0.9980	0.0380	0.9989
NasNet	0.0313	0.9941	0.0270	0.9960

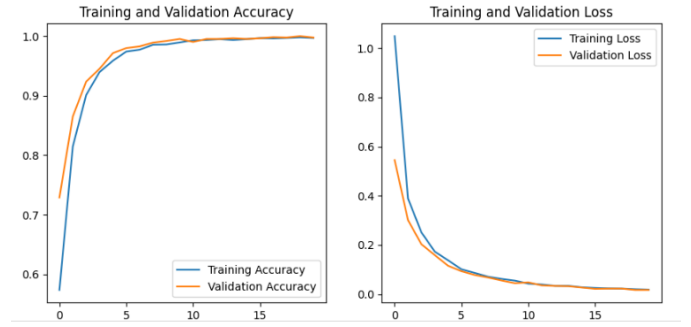
loss values of different pre-trained models when tested on the original ALL IDB Dataset, original ALL IDB Dataset + Augmentation and the generated SGALL dataset, respectively. From the tabular data analysis it is observed that, the pre-trained models demonstrated impressive accuracy when tested with original ALL IDB dataset, showcasing their capability to learn from the original data distribution. Moreover, upon augmenting the ALL IDB Dataset with additional data, these models exhibited enhanced validation accuracy, demonstrating their resilience and ability to generalize well to diverse

TABLE V: Comparison Of Accuracy, Precision, Recall, F1-Score of Different Pretrained Models using Proposed SGALL Dataset

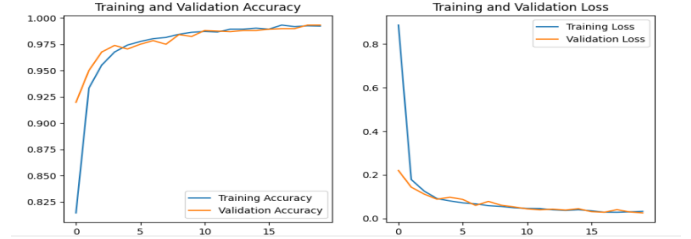
Model	Validation Accuracy	Precision	Recall	F1-score
VGG16	0.9972	0.9933	0.9929	0.9931
VGG19	0.9977	1.000	1.000	1.000
InceptionV3	0.9932	0.9824	0.9813	0.9819
Resnet50	0.9937	0.9944	0.9941	0.9942
DenseNet121	0.9977	0.9977	0.9976	0.9977
DenseNet169	0.9994	1.000	1.000	1.000
DenseNet201	0.9990	0.9988	0.9988	0.9988
EfficientNetB1	0.9989	0.9988	0.9988	0.9990
NasNet	0.9960	0.9944	0.9941	0.99

TABLE VI: Accuracy Comparison of DenseNet169 across three datasets

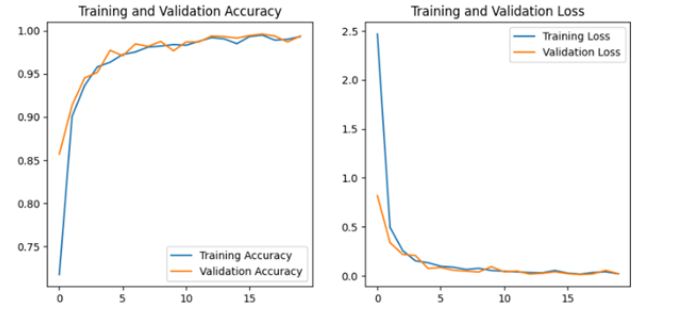
Model	ALL IDB dataset	ALL IDB dataset + Augmentation	SGALL dataset
DenseNet169	96.80%	96.82%	99.94%



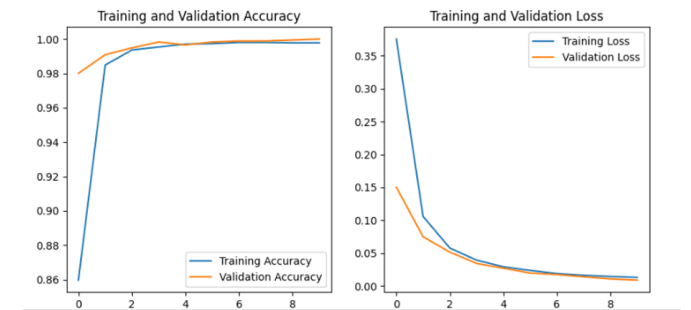
(a) Accuracy and Loss graph for VGG19



(b) Accuracy and Loss graph for INCEPTIONV3



(c) Accuracy and Loss graph for Resnet50



(d) Accuracy and Loss graph for DenseNet201

Fig. 5: Accuracy and loss curves of different Pre-trained models

data variations. However, these models when tested on our generated SGALL dataset achieved outstanding validation accuracy, substantiating their efficacy in handling synthetic data and novel image representations. Table V depicts the performance measure of compared pre-trained models when tested on SGALL dataset. Among the compared pre-trained models, the DenseNet-169 achieved the highest validation

accuracy across all datasets, further solidifying its effectiveness and adaptability across diverse data settings. To have a fair comparison, Figure 5 depicts the accuracy and loss curves of the compared models. From the plot it is observed that DenseNet169 achieved stable training behaviour when trained and validated on SGALL dataset. Table VI illustrates a comprehensive comparison of the validation accuracy achieved by the DenseNet-169 model across the compared datasets and the model outperforms when trained and tested on the generated SGALL dataset. Notably, the DenseNet-169 model with the SGALL dataset achieved 3.12% improved accuracy than the model with the original ALL IDB dataset and the ALL IDB dataset + Augmented dataset. Dense connections enable the preservation of the diversified contextual features in the generated blood smear images.

B. Performance comparison of DenseNet169 model with existing the state-of-the-art methods

This section compares the performance of the DenseNet-169 model on the SGALL dataset with some existing GAN-based DL models [13] and [8]. The comparison results are represented in Table VII.

TABLE VII: Comparison of Accuracy, Precision, Recall, F1-score of compared state-of-the-art GAN models

Model	Accuracy	Precision	Recall	F1-score
[13]	0.9989	0.9977	0.9970	0.9980
[8]	0.8610	0.8650	0.8610	0.8630
SinGAN (Proposed)	0.9994	1.000	1.000	1.000

Compared with the state-of-the-art model, our proposed model achieves not only superior accuracy but also higher precision and recall. This is because the methods in [13] and [8] utilise GANs or their variants to increase the dataset volume and, consequently, achieve better performance than conventional data augmentation techniques. However, the generated sample is not properly validated for realism, resulting in lower precision and recall. However, our proposed model validates the realism of the GAN-generated samples before their further use and thus provides the highest accuracy, precision, and recall values. These improved results highlight the efficacy and robustness of our generated dataset, showcasing its potential to revolutionise image recognition tasks and enhance overall performance in real-world applications.

V. CONCLUSION

Obtaining labelled data is time-intensive and expensive process, especially in domains that require expert annotations or specialized equipment. Data synthesis methods like GANs offer an efficient alternative by synthetically generating additional samples without the need for extensive data collection. However, the synthetically generated samples must be validated before their use in downstream applications. This paper used a lightweight GAN model, i.e., SinGAN, to generate diversified synthetic blood smear data, and the generated samples were validated using widely used annotation algorithms.

Ambiguous samples are discarded from the dataset to improve the reliability of the remaining samples. DenseNet 169 model trained and tested on the validated GAN samples achieved 99.94% accuracy and 100% precision in classifying ALL samples. In further research, SinGAN variants can facilitate transfer learning and domain adaptation, enabling the model to generalise better across different datasets and tasks.

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